

# Wookyoung Kim

19 papers SCI + KCI peer-reviewed    18 patents domestic + 1 US    8 programs registered software    5 transfers technology transfers

## Summary

Thermal engineer specializing in data center cooling, heat exchangers, heat pump systems, and two-phase heat transfer. Experienced in both experimental research and engineering software development.

## Education

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|----------------------------------------------------------------------------------------------------------------------|-------------|
| Ph.D. · KAIST (Korea Advanced Institute of Science and Technology) · Mechanical Engineering                          | 2017 – 2021 |
| · Dissertation: Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes |             |
| · Advisor: Prof. Sung Jin Kim                                                                                        |             |
| M.S. · KAIST · Mechanical Engineering                                                                                | 2015 – 2017 |
| · Thesis: Effect of artificial cavities on the thermal performance of pulsating heat pipes                           |             |
| · Advisor: Prof. Sung Jin Kim                                                                                        |             |
| B.S. · KAIST · Mechanical Engineering                                                                                | 2011 – 2015 |

## Experience

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|----------------------------------------------------------------------------------------------------------------------------------|----------------|
| Senior Researcher · Korea Institute of Machinery and Materials (KIMM) · Daejeon, Republic of Korea                               | 2021 – Present |
| Heat Pump Research Center, Department of Thermal Energy Solutions, Innovative Energy Machinery Research Division                 |                |
| – PI: Jet-enhanced immersion cooling for next-gen data center cooling                                                            |                |
| – PI: KIMMPROP - Thermophysical property iOS/Android app development                                                             |                |
| – Lead: PCHE design & testing for liquid hydrogen vaporizers (down to -220°C, 100 MPa)                                           |                |
| – Lead: Immersion cooling waste heat utilization and thermal management                                                          |                |
| – 19 journal papers (11 SCI + 8 KCI), 18 domestic patents, 1 U.S. patent, 8 registered software programs, 5 technology transfers |                |

## Publications

11 SCI · 8 KCI · 11 Conference

SCI / International Journal Articles 11 papers

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|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Falling film evaporation of ammonia on smooth and Low-Fin tube arrays<br><i>Thermal Science and Engineering Progress</i><br>Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young                                                          |
| 2025 | Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger<br><i>Applied Thermal Engineering</i><br>Kim, Bogyom and Lee, Kong Hoon and Kim, Minchang and Moon, Sunyoung and Yoo, Jin Woo and Kim, Wookyoung |
| 2024 | Experimental studies on the VLE of R-32/R-125 and verification of experimental setup<br><i>Journal of Mechanical Science and Technology</i><br>Kim, Dong Ho and Kim, Wookyoung and Kim, Jeong Hyun and Kim, Hak Soo and Lee, Kong Hoon               |

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | A numerical study on the performance of chemisorption heat pump according to various design conditions<br><i>Applied Thermal Engineering</i><br>Kim, Hak Soo and Kim, Dong Ho and Kim, Jin Sub and Kim, Wookyoung and Kim, Young                                                    |
| 2024 | Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures<br><i>Applied Thermal Engineering</i><br>Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young                                                                                   |
| 2024 | Experimental study of a chemisorption heat pump under different operation conditions<br><i>Applied Thermal Engineering</i><br>Kim, Hak Soo and Kim, Jeong Hyun and Kim, Jin Sub and Kim, Wookyoung and Kim, Young                                                                   |
| 2023 | Characterization of liquid behavior in distributor of falling-film evaporator<br><i>Physics of Fluids</i><br>Kim, Jungchul and Kim, Chan Geun and Kim, Hak Soo and Kim, Wookyoung and Kim, Dong Ho                                                                                  |
| 2023 | Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement<br><i>International Communications in Heat and Mass Transfer</i><br>Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young and Yoon, Seok Ho |
| 2021 | Fundamental issues and technical problems about pulsating heat pipes<br><i>Journal of Heat Transfer - Transactions of the ASME</i><br>Kim, Wookyoung and Kim, Sung Jin                                                                                                              |
| 2020 | Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes<br><i>International Journal of Heat and Mass Transfer</i><br>Kim, Wookyoung and Kim, Sung Jin                                                                               |
| 2018 | Effect of reentrant cavities on the thermal performance of a pulsating heat pipe<br><i>Applied Thermal Engineering</i><br>Kim, Wookyoung and Kim, Sung Jin                                                                                                                          |

#### KCI / Korean Journal Articles 8 papers

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | A Study on the Design of an Air-to-Air Plate Heat Exchanger for 300°C High Temperature Heat Pumps<br><i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i><br>Kim, Jin Sub and Shin, Dong Hwan and Shim, Jaechwan and Lee, Kong Hoon and Sohn, Sangho and Kim, Wookyoung and Song, Chanhoo        |
| 2024 | Experimental Investigation on the Freezing Condition of Printed Circuit Heat Exchanger for Cryogenic Liquid Hydrogen Vaporizer<br><i>Journal of Hydrogen and New Energy</i><br>Kim, Wookyoung and Kim, Bokyem and Sohn, Sangho and Lee, Kong Hoon and Kim, Jungchul                                                 |
| 2024 | An Experimental Study on The Thermal Performance Measurement and Flow Visualization of Falling Film Evaporator Using R-1233ZD(E) Refrigerant<br><i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i><br>Kim, Wookyoung and Kim, Hak Soo and Kim, Jungchul and Kim, Dong Ho                       |
| 2023 | An Experimental Study on the Performance Characteristics of a Falling Film Evaporator for R-1234ze(E) Refrigerant<br><i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i><br>Kim, Dong Ho and Kim, Hak Soo and Kim, Jungchul and Kim, Wookyoung                                                  |
| 2023 | Comparative Study of Air Cooling and Immersion Cooling for the Thermal Management of a Cylindrical Battery Pack<br><i>Transactions of the Korean Society of Mechanical Engineers B</i><br>Kim, Jin Sub and Kim, Seul Ah and Shin, Dong Hwan and Kim, Wookyoung and Moon, Sunyoung and Chung, Yoong and Sohn, Sangho |

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|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022                             | A Study on Anti-Icing Design by Conjugate Heat Transfer Analysis in a Lab-Scale Printed Circuit Heat Exchanger for Supply of Cryogenic High Pressure Liquid Hydrogen<br><i>Transactions of the Korean Hydrogen and New Energy Society</i><br>Sohn, Sangho and Kim, Wookyong         |
| 2022                             | Numerical Study on the Performance of Hybrid Falling Film Evaporator<br><i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i><br>Kim, Hak Soo and Kim, Wookyong and Lee, Kong Hoon and Kim, Dong Ho                                                               |
| 2022                             | Selection of the Equation of State for Thermodynamic Properties in the Low GWP Mixture Refrigerants<br><i>Transactions of the Korean Society of Mechanical Engineers B</i><br>Lee, Kong Hoon and Kim, Dong Ho and Kim, Wookyong and Kim, Jin Sub and Kim, Young and Yoo, Jin Woo    |
| Conference Proceedings 11 papers |                                                                                                                                                                                                                                                                                     |
| 2024                             | Experimental and theoretical investigation on avoiding freezing phenomena of Printed Circuit Heat Exchanger for cryogenic liquid hydrogen vaporizer<br><i>Proceedings of the 29th International Cryogenic Engineering Conference (ICEC29)</i><br>Kim, Wookyong and others           |
| 2024                             | Understanding the thermal characteristics of falling film evaporation of R-1233ZD(E) refrigerant using flow visualization and heat transfer measurement<br><i>Proceedings of the 11th Asian Conference on Refrigeration and Air-conditioning (ACRA)</i><br>Kim, Wookyong and others |
| 2023                             | Experimental investigation on the flow and thermal characteristics of falling film evaporator using R-1233ZD(E) refrigerant<br><i>Proceedings of the 11th International Conference on Boiling and Condensation Heat Transfer (ICBCHT)</i><br>Kim, Wookyong and others               |
| 2019                             | Experimental Investigation on the Thermal Performance of PHP Operating in a Circulation/oscillation Mode<br><i>Proceedings of the Korea-China-Japan Joint Symposium</i><br>Kim, Wookyong and Kim, Sung Jin                                                                          |
| 2019                             | Comparison of thermal performance between CEPHP and CLPHP<br><i>Proceedings of the 4th Thermal and Fluids Engineering Conference (TFEC)</i><br>Kim, Wookyong and Kim, Sung Jin                                                                                                      |
| 2018                             | Experimental investigation on the thermal performance of double-condenser pulsating heat pipes<br><i>Proceedings of the 10th International Conference on Boiling and Condensation Heat Transfer (ICBCHT)</i><br>Kim, Wookyong and Kim, Sung Jin                                     |
| 2018                             | Development of deployable radiator for spacecraft using a pulsating heat pipe<br><i>Proceedings of the KSME 2018 Spring Annual Meeting (Thermal Engineering Division)</i><br>Kim, Wookyong and Kim, Sung Jin                                                                        |
| 2018                             | Effects of circulating flow on the thermal performance of a pulsating heat pipe<br><i>Proceedings of the KSME 2018 Annual Meeting</i><br>Kim, Wookyong and Kim, Sung Jin                                                                                                            |
| 2017                             | Experimental Investigation on the Thermal Performance of a Pulsating Heat Pipe with Artificial Cavities<br><i>Proceedings of the 1st Asian Conference on Thermal Sciences (ACTS 2017)</i><br>Kim, Wookyong and Kim, Sung Jin                                                        |
| 2017                             | Experimental investigation of artificial cavities on the thermal performance of a pulsating heat pipe<br><i>Proceedings of the ASME InterPACK 2017</i><br>Kim, Wookyong and Kim, Sung Jin                                                                                           |
| 2016                             | Effects of artificial cavities on the startup performance of a pulsating heat pipe<br><i>Proceedings of the KSME 2016 Annual Meeting</i><br>Kim, Wookyong and Kim, Sung Jin                                                                                                         |

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## Development of an AI-based Automated Design Tool for Data Center Direct Liquid Cooling (DLC) Systems

2026

AI-driven automated design tool for data center direct liquid cooling (DLC) systems, using surrogate modeling and optimization for cold-plate and cooling-system design (PI)

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## Development of a Direct Liquid Cooling (DLC) System for Reducing Power Consumption in Data Centers

2026 – 2029

National R&D project (KETEP, Ministry of Climate, Energy and Environment) to develop a direct liquid cooling (DLC) system that reduces data center power consumption (Lead)

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## Jet-Enhanced Immersion Cooling for Data Centers

2025

Development of high-efficiency jet-impingement immersion cooling technology for next-generation high-heat-density data center servers (PI)

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## KIMMPROP - Thermophysical Property iOS/Android App

2025

Cross-platform iOS/Android app for thermophysical property calculations using CoolProp/REFPROP via WASM (PI)

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## PCHE for Liquid Hydrogen Supply System

2021 – 2026

Core material development for liquid hydrogen supply system - PCHE performance testing and design (Lead)

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## Compact PCHE Design for Liquid Hydrogen

2022 – 2026

Compact heat exchanger (PCHE) design technology for liquid hydrogen below -200°C, 100 MPa class (Lead)

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## Immersion Cooling Waste Heat Utilization

2024 – 2028

Immersion cooling waste heat utilization and active thermal management technology (Lead)

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## 300°C High-Temperature Heat Pump System

2023 – 2028

300°C-class high-temperature heat pump system for fossil fuel replacement (Participant)

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## Chemisorption Heat Pump with Electrochemical Compressor

2021 – 2025

Chemisorption heat pump system using electrochemical compressor (Participant)

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## Next-Gen Alternative Refrigerant & HVAC Technology

2021 – 2025

Next-generation alternative refrigerant and high-efficiency HVAC core technology development (Participant)

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## Low-Charge Industrial Heat Pump

2022 – 2023

Industrial heat pump with low refrigerant charge and simultaneous hot/cold water production (Participant)

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## Water-Source Heating/Cooling Hybrid System

2022 – 2023

Water-source heating/cooling and regenerative heat hybrid system (Participant)

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## Heat Pipe HX Performance Enhancement

2022

Heat pipe heat exchanger performance enhancement using boiling (Participant)

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## Post H-class Gas Turbine Technology

2022

Post H-class gas turbine technology development (Participant)

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## Compact Inverter Compressor Dress Care System

2022

Compact inverter compressor-based dress care system development (Participant)

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Ultra-Thin Vapor Chamber Performance Enhancement                                                                                                                     | 2022 – 2023 |
| Performance enhancement technology for ultra-thin vapor chambers for semiconductor cooling (Participant)                                                             |             |
| PCHE Freezing Test for Vehicle Heat Exchangers                                                                                                                       | 2025        |
| PCHE freezing performance test for automotive heat exchanger applications (PI)                                                                                       |             |
| BTMS Integrated A/Con System Evaluation                                                                                                                              | 2025        |
| Battery thermal management system integrated A/Con system real-vehicle evaluation process development (Participant)                                                  |             |
| 3D Multiporous Cooling System for Ultra-high Heat Flux Applications                                                                                                  | 2021        |
| Development of a 3D multi-porous cooling device for ultra-high-heat-flux heat-generating elements (Graduate Researcher)                                              |             |
| Smart protective suit with active cooling/ventilation and antivirus suit changing system                                                                             | 2020        |
| Active cooling and ventilation system for a smart protective suit enabling on-body thermal management (Graduate Researcher)                                          |             |
| Next-Generation CPI Reliability Structural & Thermal Design                                                                                                          | 2020        |
| Structural and thermal design technology to secure the reliability of next-generation CPI; NRF original-technology development project (Graduate Researcher)         |             |
| OHP High-Pressure Working-Fluid Charging Technology                                                                                                                  | 2020        |
| Technical service for high-pressure working-fluid charging of oscillating heat pipes (OHP); commissioned by Korea Institute of Energy Research (Graduate Researcher) |             |
| Thermal optimization of MMU rear housing for reducing the weight                                                                                                     | 2019 – 2020 |
| Heat-dissipation performance optimization and weight reduction of an MMU rear housing; industry project funded by Samsung Electronics (Graduate Researcher)          |             |
| Development of a Flexible Thin-Film Ultra-Thermal-Conductor                                                                                                          | 2015 – 2020 |
| NRF basic research on flexible thin-film ultra-high-thermal-conductivity heat spreaders based on pulsating heat pipes (Graduate Researcher)                          |             |
| Study on a smart space thermal management system                                                                                                                     | 2016 – 2018 |
| Pulsating-heat-pipe-based deployable radiator and smart thermal control device for spacecraft; contract with Korea Aerospace University (Graduate Researcher)        |             |

## | Patents & Registered Software

Patents 19 registered

|      |                                                   |
|------|---------------------------------------------------|
| 2023 | Heat exchanger with anti-freezing capability (1)  |
| 2023 | Heat exchanger with anti-freezing capability (2)  |
| 2024 | Modular fin-tube reactor                          |
| 2024 | Fin-tube type reactor                             |
| 2022 | Tray structure for heat exchanger                 |
| 2022 | Oil recovery system for refrigeration systems (1) |
| 2022 | Oil recovery system for refrigeration systems (2) |
| 2023 | Micro-channel reactor                             |
| 2024 | Immersion cooling device                          |
| 2023 | Geothermal heat supply device and heating system  |

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|------|---------------------------------------------------------------------------|
| 2023 | Heat exchanger thermal performance testing apparatus                      |
| 2022 | Ternary refrigerant composition and heat pump system                      |
| 2023 | Display device                                                            |
| 2022 | Duct assembly and combustor                                               |
| 2022 | Immersion cooling HVAC system and method                                  |
| 2022 | Heat pipe integrated reactor for adsorption heat pump                     |
| 2023 | Adsorption heat pump evaporator and system                                |
| 2018 | Plate pulsating heat spreader with artificial cavities                    |
| 2019 | Plate pulsating heat spreader with artificial cavities (US 10,436,520 B2) |

Registered Software 8 programs

|      |                                                                 |
|------|-----------------------------------------------------------------|
| 2024 | PCHE thickness design program (C-2024-052179)                   |
| 2024 | High-temperature heat pump cycle design program (C-2024-038691) |
| 2023 | Heat exchanger HTC uncertainty analysis program (C-2023-059071) |
| 2023 | Lab-scale heat exchanger analysis program (C-2023-056463)       |
| 2025 | 100 kg/hr hydrogen PCHE design program (C-2025-056891)          |
| 2025 | Adsorption cooling simulation program (C-2025-034652)           |
| 2024 | Vapor chamber performance analysis program (C-2024-041200)      |
| 2024 | Fin-tube heat exchanger design program (C-2024-038692)          |

## Core Skills

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| EXPERIMENTAL               | Two-phase heat transfer testing · Cryogenic systems (down to -220°C) · High-pressure testing (up to 100 MPa) · PCHE diffusion bonding & fabrication · Immersion cooling (CPU/GPU servers) · Flow visualization (high-speed camera) · VLE measurement & gas chromatography |
| ANALYTICAL & COMPUTATIONAL | Thermal network modeling · Heat exchanger design (PCHE, S&T, PHE) · CFD simulation · Surrogate modeling & design optimization · CoolProp/REFPROP                                                                                                                          |
| SOFTWARE DEVELOPMENT       | Python, JavaScript, TypeScript · FastAPI, React Native · Git/GitHub, Linux, CMake · Cross-platform app development (iOS/Android) · AI/LLM agent development                                                                                                               |

## Research Interests

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|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Data Center Cooling<br>Immersion cooling · Direct liquid cooling · Jet impingement · PUE optimization      | High-Heat-Flux Electronics Cooling<br>CPU/GPU cold plate · Focused cooling · Next-gen chip thermal management |
| Heat Exchangers<br>PCHE · Shell & Tube · Plate HX · Cryogenic applications                                 | Heat Pump Systems<br>Chemisorption · Adsorption · High-temperature (300°C)                                    |
| Low GWP Refrigerant Systems<br>Eco-friendly refrigerants · VLE measurement · Equation of state development | Two-Phase Heat Transfer<br>Boiling · Evaporation · Pulsating heat pipes · Vapor chambers                      |
| Cryogenic Heat Transfer<br>Liquid hydrogen vaporizer · Supercritical fluid · Sub-200°C systems             |                                                                                                               |

## Professional Affiliations

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|------------------------------------------------------------------------------------------------------------|-------------------|
| Council Member · Korean Society of Mechanical Engineers (KSME)                                             | 2026-04 – Present |
| Committee Member - Thermal Management / Cooling · SAREK Data Center Facility Technology Division Committee | 2026-03 – Present |
| Regular Member · Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK)                  | 2022-04 – Present |
| Regular Member · Korean Society of Mechanical Engineers (KSME)                                             | 2021-05 – Present |

Updated 2026-08